SOCIOL 723: Social Statistics II

Week 9, Regression Discontinuity Designs

Wenhao Jiang
Department of Sociology, Fall 2026

Duke
UNIVERSITY

Today

Sharp Regression Discontinuity

Estimation

Covariates and Power

Validity

Local Randomization

Fuzzy RD

Application: Bombing in Vietnam

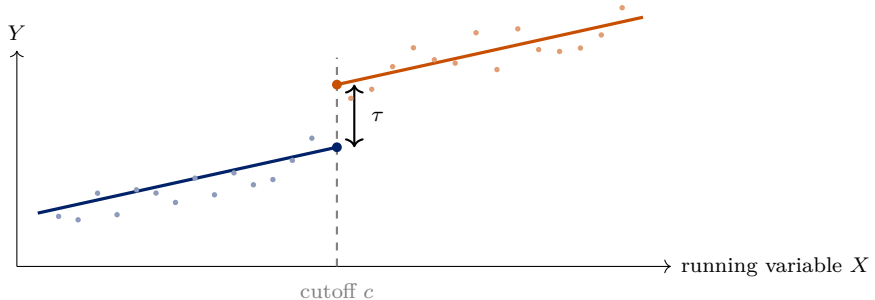
Extensions

Practice

★ = essential, you must understand this ★ = for understanding, not examined

The Idea in One Picture ★

Many treatments are assigned by a rule: a test score threshold, an income cut-off, a vote share above 50%, an age of eligibility.



Just below and just above the cutoff, units are essentially identical, except that one group got the treatment. The jump in the outcome at c is the effect.

Where RD Sits

Goal: an effect estimate when treatment is assigned by a rule on a measured score, with no overlap anywhere.

design	comparable units	needs
matching, weighting (Week 6)	same X, both treatments observed	overlap, $0 < e(x) < 1$
IV (Week 7)	moved by Z, which moves only D	exclusion, monotonicity
RD (today)	just either side of a cutoff in X	continuity at the cutoff

Plan:

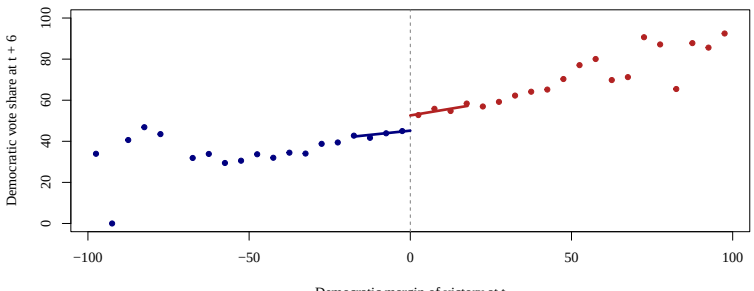
1. Sharp RD: why continuity identifies the effect at c, and only there.
2. Estimation as four choices: shape, weights, bandwidth, inference.
3. Covariates and power; validity checks; local randomization (with Dutch science grants).
4. Fuzzy RD as local IV; Dell and Querubín (2018): thresholds and a boundary.
5. Extensions: RD in time (police shootings), kinks, repeated cutoffs.

Sharp RD

Running Example: the U.S. Senate

Does winning a Senate seat help the party keep it six years later? (Lee 2008 for the House; Cattaneo, Frandsen, and Titiunik 2015 for the Senate)

- X_i: Democratic margin of victory at t, in points ($c = 0$); Y_i: Democratic vote share in the same seat at $t + 6$.
- 1,297 elections, 1914–2010 (595 lost, 702 won).



Dots: means in 5-point bins. Lines: the local linear fits used below (figs/w9_senate.R).

Winners vs Losers, in Closer and Closer Races

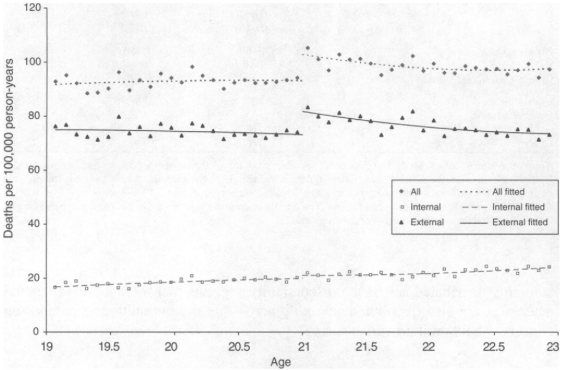
Democratic vote share at $t + 6$, by result at t (figs/w9_estimation.R):

races decided by less than	lost	won	mean, lost	mean, won	difference
any margin (all races)	595	702	40.9	62.6	21.7
50 points	565	562	41.3	57.4	16.1
20 points	389	346	43.6	55.0	11.3
10 points	245	206	44.5	54.1	9.6
5 points	128	117	45.0	52.8	7.8
2 points	49	47	42.7	52.9	10.2

- **All races**: seats Democrats won were already Democratic; their vote share six years *earlier* was 14.6 points higher. The 21.7 mixes the effect of winning with party strength.
- **Close races**: a 1-point win and a 1-point loss happen in similar places; the gap shrinks. But few races remain, and it bounces.
- Needed: why close races are comparable, and how to use them.

A Second Example: Turning 21 (Carpenter and Dobkin 2009)

Does drinking kill young adults? U.S. alcohol becomes legal on the 21st birthday.
Running variable: age in days; cutoff: 21.



Their Figure 3: U.S. deaths 1997–2004 per 100,000 person-years, 30-day bins.

Age 20 and 364 days vs. 21 and 1 day: alike in almost everything *except* legal alcohol.

Turning 21: the Jumps

Jump at 21 (Carpenter and Dobkin 2009, Tables 2, 4, 5):

outcome	jump at 21	(se)
share of days drinking (8.5% of days at age 20)	+1.8 points, +21%	(0.95)
deaths, all causes	+8.7%	(1.7)
motor vehicle accidents	+14.3%	(3.2)
suicides	+15.4%	(5.9)
deaths mentioning alcohol	+34.6%	(11.6)
homicides	+0.2%	(4.5)
internal causes (illness)	+5.4%	(4.0)

- About 10 more deaths per 100,000 person-years, mostly accidents and injuries.
- Sex, race, schooling, employment, and insurance show no jump at 21.
- The law moves drinking; dividing the death jump by the drinking jump is a **fuzzy RD** (later today).

Setup and Identification ★

A running variable (or forcing variable) X_i determines treatment deterministically:

$$D_i = \mathbb{1}\{X_i \geq c\}.$$

Write $m_d(x) = E[Y_i(d) \mid X_i = x]$ for the two potential-outcome curves.

Continuity assumption (Hahn, Todd, and van der Klaauw 2001)

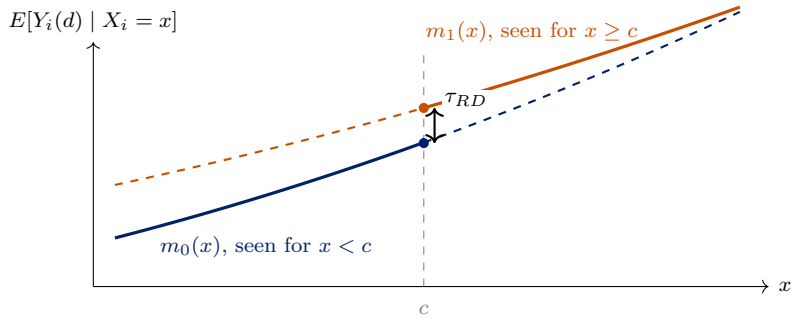
$m_1(x)$ and $m_0(x)$ are continuous in x at $x = c$.

Then

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x] = E\big[Y_i(1) - Y_i(0) \mid X_i = c\big].$$

- Not assumed: anything about the shape of $E[Y \mid X]$ away from the cutoff, or ignorability anywhere else.
- Positivity fails at every x: $\Pr(D = 1 \mid X = x)$ is 0 or 1. Identification comes from continuity, not overlap, which is why the estimand is local to c.

Two Curves, Each Seen on One Side



- The observed regression $E[Y_i \mid X_i = x]$ follows m_0 to the left of c and m_1 to the right; it jumps at c by $m_1(c) - m_0(c)$.
- The dashed halves are never observed. Away from c the gap between the curves, $\tau(x)$, is not identified without further assumptions.

What Continuity Rules Out

Continuity fails when something other than D changes at c.

- **A second treatment at the same cutoff.** An age threshold that grants a pension and also lowers a tax; a district boundary where schools, police, and taxes all change. The jump is the effect of the bundle.
- **Sorting.** If people who would do well anyway can place themselves just above c, the units just above differ from those just below in $Y_i(0)$ itself, and m_0 jumps.
- **Selection of the cutoff on outcomes.** An agency that sets c where it expects a change, after seeing the data, builds the jump in.

Continuity is a statement about $Y_i(0)$ and $Y_i(1)$, so it cannot be tested directly. It has two testable implications: the density of X and the distribution of pre-determined covariates should not jump at c.

The Estimand Is Local, and That Matters ★

τ_{RD} is the average effect for units at the cutoff. Nothing more.

- A scholarship RD tells you about students at the eligibility margin, not about the strongest or weakest applicants.
- Extrapolating away from c requires a functional-form assumption of exactly the kind RD was designed to avoid.
- Compare with the LATE from Week 7: both are estimands you did not choose, determined by the design rather than by the question. State them plainly.
- In exchange, RD is often the most credible observational design there is. The identifying variation is visible in a scatterplot.

Lee (2008)'s reformulation is helpful: if agents cannot perfectly control X_i, then treatment near the cutoff is *as good as randomly assigned*, and the design is a local experiment.

Identification Is Done. What Is Left? ★

Target: $\tau_{RD} = m_1(c) - m_0(c)$, the heights of the two curves *exactly at* the cutoff.

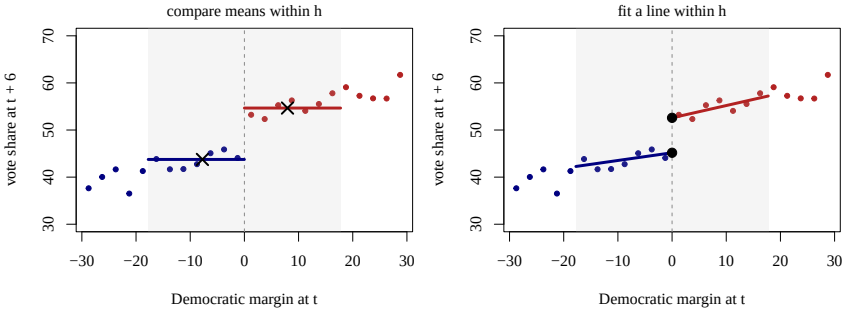
Problem: no unit sits exactly at c (Senate: no race is an exact tie; 23 of 1,297 are within half a point). Each height must be predicted from units near c. Four choices, each shown first on the Senate data:

choice	question	what is at stake	standard answer
shape	Mean, line, or curve?	mean: biased if the curve slopes; curve: more variance	a line on each side
weights	Closer units count more?	far units add bias	triangular kernel
bandwidth	How far from c?	wider: less variance, more bias	minimize MSE
inference	How sure?	leftover bias: usual interval too short	robust bias-corrected

Estimation

Shape: Two Ways to Use the Races Near the Cutoff ★

Keep the 683 races within $h = 17.75$ points of a tie (shaded) (figs/w9_estimation.R):



- **Left**: average each side. $54.66 - 43.75 = 10.91$.
- **Right**: fit a line on each side and read off its height at margin 0 (black dots). $52.59 - 45.17 = 7.41$.

How Much Comparing Means Overstates: the Algebra

Goal: write the difference in means as the line estimate plus an error.

Fact: an OLS line passes through the point of means $(\bar{x}, \bar{Y})$. The normal equation for the intercept says the residuals sum to zero:

$$\textstyle\sum_i (Y_i - \hat{\mu} - \hat{\beta} x_i) = 0 \quad \Rightarrow \quad \bar{Y} = \hat{\mu} + \hat{\beta}\, \bar{x}.$$

Apply it on each side, with $x_i = X_i - c$ and $\bar{x}_\pm$ the mean distance (uniform kernel):

$$\begin{aligned}
\bar{Y}_+ &= \hat{\mu}_+ + \hat{\beta}_+ \bar{x}_+ && \text{the line above passes through its means} \\
\bar{Y}_- &= \hat{\mu}_- + \hat{\beta}_- \bar{x}_- && \text{the line below, likewise} \\
\bar{Y}_+ - \bar{Y}_- &= \underbrace{(\hat{\mu}_+ - \hat{\mu}_-)}_{\hat{\tau}_{\text{linear}}} + \hat{\beta}_+ \bar{x}_+ - \hat{\beta}_- \bar{x}_- && \text{subtract} \\
&= \hat{\tau}_{\text{linear}} + \hat{\beta}_+ \bar{x}_+ + \hat{\beta}_- |\bar{x}_-| && \bar{x}_- < 0 \text{ below the cutoff}
\end{aligned}$$

With upward slopes, both extra terms are positive: the mean above is read to the *right* of c (too high), the mean below to the *left* (too low).

How Much Comparing Means Overstates: the Senate

$h = 17.75$, uniform kernel (figs/w9_senate.R): $\bar{Y}_+ = 54.66$, $\bar{Y}_- = 43.75$; $\bar{x}_+ = 7.96$, $\bar{x}_- = -7.75$; $\hat{\beta}_+ = 0.260$, $\hat{\beta}_- = 0.226$; $\hat{\tau}_{\text{linear}} = 7.09$.

$$54.66 - 43.75 = 7.09 + 0.260 \times 7.96 + 0.226 \times 7.75 \qquad \text{previous frame}$$

$$10.91 = 7.09 + 2.07 + 1.75$$

How big in general? If X is spread evenly inside the window, the means sit at the middle of each half: $\bar{x}_+ \approx h/2$, $\bar{x}_- \approx -h/2$. Then

$$\bar{Y}_+ - \bar{Y}_- - \hat{\tau}_{\text{linear}} \approx (\hat{\beta}_+ + \hat{\beta}_-)\,\frac{h}{2} \qquad \text{grows in proportion to } h$$

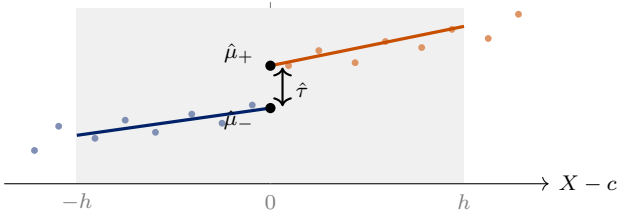
(Senate: $h/2 = 8.9$ against the actual 7.96 and 7.75.)

- The error of comparing means is of order h; here it inflates the effect by half.
- A line removes this term. What is left comes from curvature and is of order h^2 (bandwidth section).

Shape: a Line on Each Side ★

In words:

1. Keep the units within h of the cutoff.
2. On each side, regress Y on the distance to the cutoff, $X_i - c$.
3. Each intercept is the predicted Y *at* the cutoff on that side.
4. $\hat{\tau}$ is the difference between the two intercepts.



Senate: below, 45.17 at margin 0; above, 52.59; $\hat{\tau} = 7.41$ (right panel of the previous frame).

The Same Line Fits as One Regression

On the units with $|X_i - c| \le h$ (same answer as two separate fits):

$$Y_i = \mu + \tau D_i + \beta_-(X_i - c)(1 - D_i) + \beta_+(X_i - c)D_i + \text{error}$$

coefficient	meaning
μ	predicted Y just below the cutoff
$\mu + \tau$	predicted Y just above the cutoff
τ	the jump at the cutoff: the RD estimate
β_-, β_+	slopes below and above, free to differ

- Centre the running variable at c, so that τ is the gap at the cutoff rather than at $X = 0$.
- Let the slopes differ: a common slope lets data on one side shape the prediction on the other.

The Estimator on the Senate Data

`rdrobust` defaults: local linear, triangular kernel, $h = 17.75$ points (figs/w9_senate.R).

intercept $= \hat{\mu}_-$ (lost by a hair)	45.17
$D = \hat{\tau}$	7.41
slope below, $\hat{\beta}_-$	0.164
slope above, $\hat{\beta}_+ = 0.164 + 0.097$	0.261
`rdrobust` conventional estimate	7.41
effective n: below, above	360, 323

- A party that barely wins a seat gets $\hat{\mu}_+ = 52.6\%$ of the vote six years later; one that barely loses gets 45.2%.
- 683 of 1,297 elections carry positive weight; the rest do not enter the estimate.

Shape: Why a Line and Not a Curve?

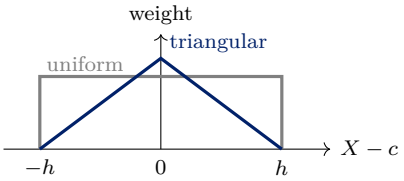
More flexible fits follow the curve better near c but are noisier. Senate, same races ($h = 17.75$, triangular weights) (figs/w9_estimation.R):

fit on each side	estimate	s.e.	problem
mean (degree 0)	9.68	0.80	biased whenever the curve slopes
line (degree 1)	7.41	1.46	biased only where the curve bends
quadratic (degree 2)	8.32	2.06	s.e. 40% larger

- A line removes the slope bias of comparing means and costs a modest amount of precision. A quadratic buys little more and costs a lot.
- The quadratic still has a job: it is how the bias of the line is estimated (inference section).

Weights: Closer Races Count More

A race won by 2 points tells us more about a tie than one won by 16. The triangular kernel weights each unit by how close it is:



$$K\left(\frac{X_i - c}{h}\right) = 1 - \frac{|X_i - c|}{h}$$

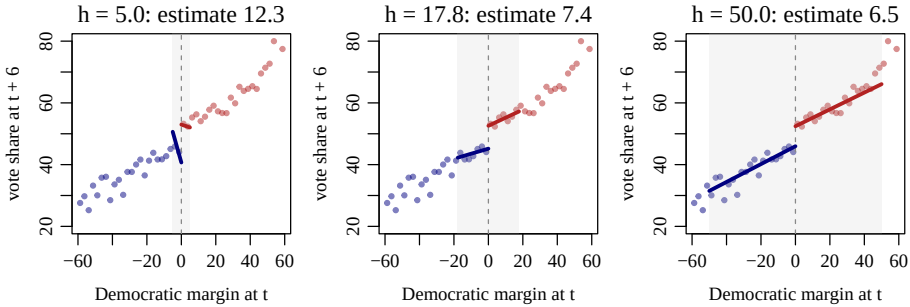
for $|X_i - c| \leq h$, zero beyond. Senate, $h = 17.75$:

- margin 2: weight 0.89
- margin 9: weight 0.49
- margin 16: weight 0.10

- Fitting the line becomes weighted least squares.
- Races within $h/2$ of a tie are 59% of the races, 80% of the weight.
- Uniform weights give 7.09, triangular 7.41. The kernel matters little; the bandwidth matters much more.

Bandwidth: How Far From the Cutoff to Look?

The same local linear fit with three bandwidths (shaded) (figs/w9_estimation.R):



Races used: 245, 683, 1,127. Standard errors: 2.55, 1.46, 1.00.

- **Narrow**: only near-ties, so the line describes the right races, but few of them. The estimate is noisy.
- **Wide**: many races and a small s.e., but one straight line must now cover landslides too. If the curve bends, the line misses at c.

Bias and Variance: Definitions ★

$\hat{\tau}_h$, the estimate at bandwidth h, varies from sample to sample:

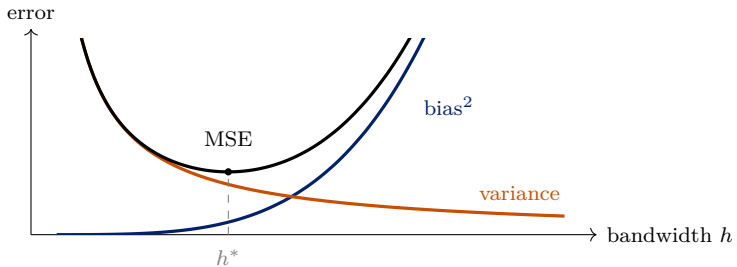
$$\begin{aligned}
\text{Bias}(\hat{\tau}_h) &= E[\hat{\tau}_h] - \tau && \text{average estimate minus the truth} \\
\text{Var}(\hat{\tau}_h) &= E\big[(\hat{\tau}_h - E[\hat{\tau}_h])^2\big] && \text{scatter around that average} \\
\text{MSE}(\hat{\tau}_h) &= E\big[(\hat{\tau}_h - \tau)^2\big] && \text{average squared distance from the truth}
\end{aligned}$$

The decomposition:

$$\begin{aligned}
\text{MSE}(\hat{\tau}_h) &= E\Big[\big((\hat{\tau}_h - E[\hat{\tau}_h]) + (E[\hat{\tau}_h] - \tau)\big)^2\Big] && \text{add and subtract } E[\hat{\tau}_h] \\
&= E\big[(\hat{\tau}_h - E[\hat{\tau}_h])^2\big] + (E[\hat{\tau}_h] - \tau)^2 && \text{expand the square;} \\
&\quad + 2\,(E[\hat{\tau}_h] - \tau)\,\underbrace{E\big[\hat{\tau}_h - E[\hat{\tau}_h]\big]}_{=0} && \text{the bias is a constant} \\
&= \text{Var}(\hat{\tau}_h) + \text{Bias}(\hat{\tau}_h)^2
\end{aligned}$$

In RD, **bias** grows with h (the line misses the curve at c) and **variance** falls with h (more units): the bandwidth trades one for the other.

The Bandwidth Is the Whole Game ★



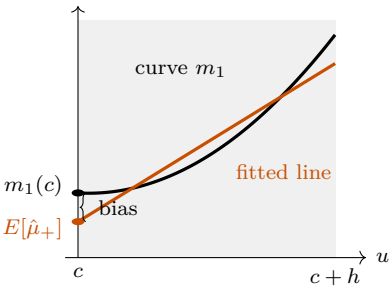
- **Wide** h: more data, lower variance, but the linear approximation is worse far from c, bias.
- **Narrow** h: less bias, but few observations, variance.
- Choose h by minimizing estimated MSE (Imbens–Kalyanaraman; the Calonico–Cattaneo–Titiunik version in `rdrobust`).
- The derivations that follow make each curve precise: bias grows like h^2, variance falls like $1/(nh)$, and h^* balances them.

The Bias of the Line, Step 1: the Picture ★

Goal: compute $\text{Bias}(\hat{\tau}) = E[\hat{\tau}] - \tau$, where $\tau = m_1(c) - m_0(c)$ and $\hat{\tau} = \hat{\mu}_+ - \hat{\mu}_-$:

$$E[\hat{\tau}] - \tau = \big(E[\hat{\mu}_+] - m_1(c)\big) - \big(E[\hat{\mu}_-] - m_0(c)\big) \qquad \text{expectation of a difference}$$

Both sides work the same way; take the side above: uniform kernel, X spread evenly, $u_i = (X_i - c)/h \in [0, 1]$. OLS is linear in Y and $E[Y_i \mid X_i] = m_1(X_i)$, so $E[\hat{\mu}_+]$ is the intercept of the line fitted to the curve m_1 itself.



- The bias is the gap at c between the curve and the line fitted to it.
- A straight curve: the line lies on it, no gap. A bent curve: the line cuts across the bend and misses at the edge.
- To compute the gap, we need a formula for m_1. We do not have one.

The Bias of the Line, Step 2: Use a Parabola

Taylor: near c, a smooth curve is close to a parabola built from its height, slope, and bend *at* c. With $X = c + hu$:

$$m_1(c + hu) \approx \underbrace{m_1(c)}_{a} + \underbrace{m_1'(c)\,h}_{b}\,u + \underbrace{\tfrac{1}{2}\,m_1''(c)\,h^2}_{q}\,u^2$$

- Why this helps: we cannot fit a line to an unknown curve by hand, but we can fit one to a parabola.
- Why expand at c: the target then appears in the formula, $a = m_1(c)$.
- Left out: $\frac{1}{6}m_1'''(c)\,h^3u^3 + \ldots$ We assume this is negligible inside the window: on $[c, c + h]$ the curve is close to a parabola.

The Bias of the Line, Step 3: Plug the Parabola Into OLS

Goal: $E[\hat{\mu}_+] - m_1(c)$. $E[\hat{\mu}_+]$ is the intercept of the OLS line of $a + bu + qu^2$ on u (Steps 1–2), with u spread evenly on $[0, 1]$.

$$\begin{aligned}
\text{slope} &= \frac{\operatorname{Cov}(u,\ a + bu + qu^2)}{\operatorname{Var}(u)} && \text{OLS slope} \\
&= \frac{b\operatorname{Var}(u) + q\operatorname{Cov}(u, u^2)}{\operatorname{Var}(u)} = b + q && a \text{ is a constant; } \operatorname{Cov}(u, u^2) = \operatorname{Var}(u) \\
E[\hat{\mu}_+] &= E[a + bu + qu^2] - (b + q)\,E[u] && \text{the OLS line passes through the means} \\
&= a + \tfrac{1}{2}b + \tfrac{1}{3}q - \tfrac{1}{2}b - \tfrac{1}{2}q = a - \tfrac{1}{6}q && E[u] = \tfrac{1}{2},\ E[u^2] = \tfrac{1}{3} \\
E[\hat{\mu}_+] - m_1(c) &= -\tfrac{1}{6}q = -\tfrac{1}{12}\,m_1''(c)\,h^2 && a = m_1(c),\ q = \tfrac{1}{2}m_1''(c)h^2
\end{aligned}$$

- b cancels: the line gets the level and the slope right. Only the bend q leaves a gap at the cutoff.

The Bias of the Line, Step 4: Both Sides Together ★

Above the cutoff (Step 3): $E[\hat{\mu}_+] - m_1(c) = -\frac{1}{12}\, m_1''(c)\, h^2$.

Below the cutoff, $u \in [-1, 0]$: the same steps give $E[\hat{\mu}_-] - m_0(c) = -\frac{1}{12}\, m_0''(c)\, h^2$.

$$\begin{aligned} \text{Bias}(\hat{\tau}) &= \big(E[\hat{\mu}_+] - m_1(c)\big) - \big(E[\hat{\mu}_-] - m_0(c)\big) && \text{Step 1} \\ &= \underbrace{-\tfrac{1}{12}\big(m_1''(c) - m_0''(c)\big)}_{B}\, h^2 \end{aligned}$$

- Bias $= Bh^2$: doubling h multiplies it by 4.
- Straight curves ($m'' = 0$) give no bias; equal bends on the two sides cancel.
- A triangular kernel or uneven X changes only the constant $\frac{1}{12}$, not the form Bh^2.

The Variance, Step 1: The Mean and the Slope

Goal: $\mathrm{Var}(\hat{\mu}_+)$, with m units above c and u spread evenly on $[0,1]$. **Assume** $Y_i = m_1(X_i) + \epsilon_i$, independent errors, $\mathrm{Var}(Y_i \mid X_i) = \sigma_+^2$. Given $\boldsymbol{X} = (X_1, \ldots, X_m)$, $m_1(X_i)$ and u_i are fixed; only the ϵ_i are random.

$$\mathrm{Var}(\bar{Y} \mid \boldsymbol{X}) = \mathrm{Var}\left(\tfrac{1}{m}\textstyle\sum_i \epsilon_i \;\middle|\; \boldsymbol{X}\right) = \dfrac{\sigma_+^2}{m} \qquad \text{drop the fixed } m_1(X_i)\text{; independence}$$

$$\mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}) = \mathrm{Var}\left(\dfrac{\sum_i (u_i - \bar{u})\,\epsilon_i}{\sum_i (u_i - \bar{u})^2} \;\middle|\; \boldsymbol{X}\right) \qquad \text{OLS slope; drop the fixed part}$$

$$= \dfrac{\sigma_+^2 \sum_i (u_i - \bar{u})^2}{\left(\sum_i (u_i - \bar{u})^2\right)^2} = \dfrac{\sigma_+^2}{m\,\mathrm{Var}(u)} \qquad \mathrm{Var}(u) = \tfrac{1}{m}\textstyle\sum_i (u_i - \bar{u})^2$$

$$\mathrm{Cov}(\bar{Y}, \hat{\beta} \mid \boldsymbol{X}) = 0 \qquad \text{see below}$$

- A high error always raises $\bar{Y}$, but raises the slope if the unit is right of $\bar{u}$ and lowers it if left. With independent, equal-variance errors, these cancel.

The Variance, Step 2: One Side

The OLS intercept is $\hat{\mu}_+ = \bar{Y} - \hat{\beta}\,\bar{u}$, so with Step 1:

$$\begin{aligned}
\operatorname{Var}(\hat{\mu}_+ \mid \boldsymbol{X}) &= \operatorname{Var}(\bar{Y} \mid \boldsymbol{X}) + \bar{u}^2 \operatorname{Var}(\hat{\beta} \mid \boldsymbol{X}) - 2\bar{u} \operatorname{Cov}(\bar{Y}, \hat{\beta} \mid \boldsymbol{X}) && \text{variance of a difference} \\
&= \frac{\sigma_+^2}{m} + \bar{u}^2 \frac{\sigma_+^2}{m \operatorname{Var}(u)} && \text{Step 1} \\
&= \frac{\sigma_+^2}{m}\left(1 + \frac{1/4}{1/12}\right) = \frac{4\,\sigma_+^2}{m} && \bar{u} = \tfrac{1}{2},\ \operatorname{Var}(u) = \tfrac{1}{12}
\end{aligned}$$

- More noise σ_+^2: larger variance. More units m: smaller variance.
- A wider h brings in more units, so m grows; σ_+^2 is the noise around the curve and does not depend on h.

The Variance, Step 3: Both Sides Together

How many units fall in $[c, c+h]$? With f the density of X:

$$m \approx n \Pr(c \le X \le c+h) = n \int_c^{c+h} f(x)\,dx \qquad \text{expected count} = n\times \text{probability}$$

$$\approx n\, f(c)\, h \qquad f \approx f(c) \text{ on the window}$$

$$\mathrm{Var}(\hat{\mu}_+) \approx \frac{4\,\sigma_+^2}{n\, f(c)\, h} \qquad \text{Step 2, substitute } m$$

$$\mathrm{Var}(\hat{\tau}) = \mathrm{Var}(\hat{\mu}_+) + \mathrm{Var}(\hat{\mu}_-) \qquad \text{the two sides use different units}$$

$$\approx \underbrace{\frac{4\,(\sigma_+^2 + \sigma_-^2)}{f(c)}}_{V} \cdot \frac{1}{nh}$$

- Variance $= V/(nh)$: doubling h halves it.
- More noise (σ^2) or fewer units near c ($f(c)$ small): more variance.

The MSE-Optimal Bandwidth: the Derivation

Goal: the h that minimizes $\text{MSE}(h) = \underbrace{B^2h^4}_{\text{bias}^2} + \underbrace{V/(nh)}_{\text{variance}}$ (the two previous derivations).

$$\frac{d}{dh} B^2h^4 = 4B^2h^3 \qquad \text{power rule}$$

$$\frac{d}{dh}\frac{V}{nh} = \frac{V}{n}\cdot\frac{d}{dh}h^{-1} = -\frac{V}{nh^2} \qquad \frac{V}{n} \text{ is a constant}$$

$$0 = 4B^2h^3 - \frac{V}{nh^2} \qquad \text{at the minimum, the slope of MSE is zero}$$

$$4B^2h^5 = \frac{V}{n} \qquad \text{multiply by } h^2\text{, move one term across}$$

$$h^* = \left(\frac{V}{4B^2\,n}\right)^{1/5} = \left(\frac{V}{4B^2}\right)^{1/5} n^{-1/5} \qquad \text{take the fifth root}$$

The MSE-Optimal Bandwidth: Reading the Formula

$$h^* = \left(\frac{V}{4B^2}\right)^{1/5} n^{-1/5}, \qquad V = \frac{4\,(\sigma_+^2 + \sigma_-^2)}{f(c)}, \qquad B = -\tfrac{1}{12}\big(m_1''(c) - m_0''(c)\big)$$

- **Wider** when the noise σ^2 is large or few units sit near c ($f(c)$ small): more data are needed to tame the variance.
- **Narrower** when the curves bend sharply at c (B large): a line is only good close to c.
- B and V are unknown: a pilot fit estimates m'', σ^2, and $f(c)$ and plugs them in (Imbens and Kalyanaraman 2012; Calonico, Cattaneo, and Titiunik 2014).
- h^* shrinks slowly: ten times the data narrows it by a factor of $10^{1/5} = 1.6$.
- Senate: $\hat{h}^* = 17.75$ points on both sides.

Bandwidth Sensitivity on the Senate Data

Conventional estimate and robust bias-corrected 95% interval (figs/w9_senate.R):

h	estimate	robust 95% interval	effective n (below, above)
5	12.27	[7.93, 19.77]	128, 117
10	7.98	[4.39, 13.11]	245, 206
12.41 (h_{CER})	7.63	[4.07, 11.29]	284, 248
17.75 (h_{MSE})	7.41	[4.09, 10.92]	360, 323
30	7.28	[5.13, 10.71]	474, 457
50	6.48	[4.87, 9.70]	565, 562

- Stable between $h = 10$ and 50; every interval excludes zero.
- At $h = 5$ the estimate rises, with 245 elections: noise, and the interval says so.
- h_{MSE}: the MSE-optimal bandwidth. h_{CER}: the **coverage-error-optimal** bandwidth, narrower, chosen so the interval covers τ as close to 95% of the time as possible (Calonico, Cattaneo, and Farrell 2020).

Inference: the Standard Error

Goal: $\mathrm{se}(\hat{\tau})$, for the usual interval $\hat{\tau} \pm 1.96\,\mathrm{se}$. Revisit the variance:

$$\mathrm{Var}(\hat{\tau} \mid \boldsymbol{X}) = \mathrm{Var}(\hat{\mu}_+ \mid \boldsymbol{X}) + \mathrm{Var}(\hat{\mu}_- \mid \boldsymbol{X}) \qquad \text{the two sides use different units}$$

$$= \frac{4\,\sigma_+^2}{m_+} + \frac{4\,\sigma_-^2}{m_-} \qquad \text{Variance, Step 2, on each side}$$

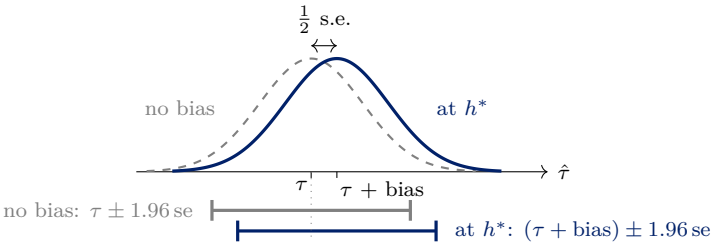
$$\widehat{\mathrm{se}}(\hat{\tau}) = \sqrt{\frac{4\,\hat{\sigma}_+^2}{m_+} + \frac{4\,\hat{\sigma}_-^2}{m_-}} \qquad \hat{\sigma}_\pm^2\text{: residual variance of each side's line}$$

Senate, $h = 17.75$ (figs/w9_estimation.R): $\hat{\sigma}_+ = 9.06$, $m_+ = 323$; $\hat{\sigma}_- = 10.20$, $m_- = 360$.

$$\widehat{\mathrm{se}}(\hat{\tau}) = \sqrt{\frac{4 \cdot 9.06^2}{323} + \frac{4 \cdot 10.20^2}{360}} = 1.47$$

- `rdrobust` reports 1.46: triangular weights, and each unit's error variance estimated from its 3 nearest neighbours in X (unequal variances allowed).
- Usual interval: $7.41 \pm 1.96 \times 1.46 = [4.56,\ 10.27]$. It ignores the bias.

Inference: How Sure? The Problem in One Picture



- At h^* the estimates average τ + bias, bias $= \frac{1}{2}$ se (next frame). So the interval is shifted up: both ends are too high.
- A shifted interval misses τ more often: it contains τ in 92% of samples, not 95%.
- Three repairs: undersmooth, robust bias correction, honest intervals.

At h^*, the Bias Is Half a Standard Error ★

Start from the first-order condition that defines h^*:

$$4B^2h^{*5} = \frac{V}{n} \qquad \text{the MSE derivation}$$

$$B^2h^{*4} = \frac{1}{4} \cdot \frac{V}{nh^*} \qquad \text{divide both sides by } 4h^*$$

$$\Big(\underbrace{Bh^{*2}}_{\text{bias}}\Big)^2 = \frac{1}{4} \quad \underbrace{\frac{V}{nh^*}}_{\text{variance}} \qquad \text{recognize the two pieces}$$

$$|\text{bias}| = \frac{1}{2}\,\text{se} \qquad \text{square root of both sides}$$

More data do not help. With $h^* \propto n^{-1/5}$: bias $= Bh^{*2} \propto n^{-2/5}$ and se $= (V/(nh^*))^{1/2} \propto (n \cdot n^{-1/5})^{-1/2} = n^{-2/5}$. Both shrink at the same rate, so their ratio stays $\frac{1}{2}$ for every n.

Three Repairs, Side by Side

Senate data (figs/w9_estimation.R, w9_senate.R):

	what it does about the bias	h	95% interval
none (usual interval)	ignores it: covers 92% at h_{MSE}	17.75	[4.56, 10.27]
1. undersmooth	shrinks it: smaller h, bias small next to the se	10	[4.38, 11.59]
2. robust bias correction	estimates it, subtracts it, widens the se	17.75	[4.09, 10.92]
...at h_{CER}	same repair, h chosen for coverage, not MSE	12.41	[4.07, 11.29]
3. honest interval ★	bounds it ($\lvert m'' \rvert \leq M$), widens by the worst case	9.84	[3.83, 12.23]

- `rdrobust`'s "robust" interval (and the sensitivity table) is repair 2. h_{CER} is not a fourth repair: it is the bandwidth for repair 2.
- Undersmoothing has no rule for how much smaller; repairs 2 and 3 do.

Repair 2: Robust Bias Correction, the Idea

The bias of the line is Bh^2, and B depends on how much the curves *bend* at c (Calonico, Cattaneo, and Titiunik 2014). So:

1. **Estimate the bend.** Fit a quadratic on each side (with a pilot bandwidth b, chosen to estimate the bend with minimum MSE; usually wider than h). Its curvature gives $\hat{B}$.
2. **Subtract the estimated bias**: $\hat{\tau}^{bc} = \hat{\tau}_h - \hat{B}h^2$.
3. **Widen the interval.** $\hat{B}$ is itself estimated with noise, so $\hat{\tau}^{bc}$ is noisier than $\hat{\tau}_h$. The robust standard error counts that extra noise.

- The robust interval is centred on $\hat{\tau}^{bc}$, not on $\hat{\tau}_h$. Senate: $\hat{\tau}_h = 7.41$, $\hat{\tau}^{bc} = 7.51$, robust interval [4.09, 10.92], whose midpoint is 7.51.
- Common practice (Cattaneo, Idrobo, and Titiunik 2020): report $\hat{\tau}_h$ (smaller MSE, since $\hat{B}$ adds noise) with the robust interval, and say so. `rdrobust` prints both rows.

Repair 2: Robust Bias Correction, the Formulas ★

Write the estimate as truth plus bias plus noise, and subtract an estimate of the bias:

$$\hat{\tau}_h = \tau + Bh^2 + \text{noise} \qquad \text{previous derivations}$$

$$\hat{\tau}^{\,bc} = \hat{\tau}_h - \hat{B}\,h^2 \qquad \hat{B}\text{: local quadratic, bandwidth } b$$

$$\hat{\tau}^{\,bc} - \tau = (B - \hat{B})\,h^2 + \text{noise} \qquad \hat{B}\text{'s error replaces the bias}$$

Its s.e. $\hat{B}$ adds noise. With $b = h$, the line minus the estimated bend is the local quadratic fit itself: same s.e. formula, a quadratic's edge factor.

$$\text{se}_{\text{robust}} = \sqrt{\frac{9\,\sigma_+^2}{m_+} + \frac{9\,\sigma_-^2}{m_-}} \qquad \text{9 for a quadratic at the edge, 4 for a line}$$

- Same σ and m: $\sqrt{9/4} = 1.5$ times the line's s.e. Senate: 2.21 (figs/w9_estimation.R); `rdrobust` (triangular weights) 2.06.
- The default b is wider than h (28.03 here): more units estimate the bend, so less extra noise; `rdrobust` gives 1.74.

Repair 2 on the Senate

`rdrobust` defaults (figs/w9_senate.R):

	estimate	s.e.	95% interval
conventional, $h = 17.75$	7.41	1.46	[4.56, 10.27]
robust bias-corrected, $b = 28.03$	7.51	1.74	[4.09, 10.92]
robust bias-corrected at $h_{CER} = 12.41$	7.68	1.84	[4.07, 11.29]

- The correction moves 7.41 to 7.51: the curves bend little near a tie.
- The robust s.e. is 19% larger (1.74/1.46): the price of estimating the bend.
- h_{MSE} is best for the *estimate*. For the *interval*, use h_{CER}, the h that makes coverage closest to 95% (Calonico, Cattaneo, and Farrell 2020). It shrinks faster with n: like $n^{-1/4}$ instead of $n^{-1/5}$. `rdrobust` rescales by the ratio of the rates, $n^{-1/4}/n^{-1/5} = n^{-1/20}$ (a rule of thumb):
  $h_{CER} = 17.75 \times 1297^{-1/20} = 17.75 \times 0.699 = 12.41$.

Does Repair 2 Work? A Simulation

The design of Calonico, Cattaneo, and Titiunik (2014): quintic curves fitted to Lee's House data, $\tau = 0.04$, 1,000 draws (figs/w9_sims.R).

coverage of a nominal 95% interval	$n = 500$	$n = 5{,}000$
conventional, MSE-optimal h	0.896	0.784
robust, MSE-optimal h	0.917	0.889
robust, CER-optimal h	0.930	0.935

- The conventional interval gets *worse* with more data.
- Practice: report the point estimate at h_{MSE} and the robust interval at h_{CER} (`bwselect = "cerrd"`).

Estimation in Practice: the Senate in Five Lines ★

1. **Plot** binned means of Y against X, with the cutoff marked. The jump should be visible before any estimate.
2. **Estimate** with a line on each side, triangular weights, h_{MSE}: $\hat{\tau} = 7.41$ points, $h = 17.75$.
3. **Interval**: robust bias-corrected, [4.09, 10.92] (at h_{CER}: [4.07, 11.29]).
4. **Report the units used**: 360 races below, 323 above.
5. **Show the choices do not drive the result**: h from 10 to 50 gives 6.5 to 8.0; a quadratic gives 8.32.

`rdrobust(y = vote, x = margin)` gives the estimate, interval, and units;
`rdplot(vote, margin)` gives the plot.

Covariates

Adding Covariates (Calonico, Cattaneo, Farrell, and Titiunik 2019)

Same local regression (units within h, kernel weights), plus covariates Z_i:

$$Y_i = \alpha + \tau D_i + \beta_-(X_i - c)(1 - D_i) + \beta_+(X_i - c)D_i + Z_i'\gamma + \epsilon_i$$

Z_i must be **pre-determined**: fixed before the cutoff is crossed. Turning 21: sex, race, schooling.

- If Z is continuous at c, it is balanced across the cutoff, so $\hat{\tau}$ should barely change. What changes is the s.e.: Z explains part of Y, so less noise is left.
- If $\hat{\tau}$ *does* change a lot, Z jumps at c: a sign of sorting, not an improvement. (Formal balance test: validity section.)
- Rules: Z enters on its own (no $D \times Z$ interaction); never a post-treatment Z (turning 21: not drinking, not a later arrest).

Covariates on the Senate Data

1,205 elections with all four covariates, triangular kernel (figs/w9_senate.R):

	jump in Z at c, $\hat{\tau}_Z$ (p)	$\tilde{\tau}$
no covariates		8.19
+ Democratic vote share, $t-6$	3.68 (0.20)	7.72
+ Democratic vote share, $t-12$	2.12 (0.34)	7.83
+ Democratic presidential vote share	−0.13 (0.95)	8.24
+ open seat	−0.14 (0.10)	8.15
all four		7.45

Every entry: `rdrobust` at h_{CER} (each fit selects its own, 12–15 points), bias-corrected estimate, robust p.

- The lagged vote jumps most (3.7 points) and moves $\tilde{\tau}$ most (8.19 → 7.72). No jump is significant at 5%.
- `rdrobust` selects its own bandwidth for each fit, with and without covariates. Report both estimates.

Covariates, Power, and What to Report

- **Covariates are for precision only.** They must be *pre-determined*, they should enter additively (not interacted with D), and the estimate should barely move when you add them. If it moves a lot, the design is in trouble, not improved (Calonico et al. 2019).
- **Effective sample size**: report the number of observations *inside the bandwidth* on each side. An RD with $n = 100{,}000$ often has 400 observations doing the work, and the standard errors should be read accordingly. Senate: 1,297 races, 683 do the work.

Validity

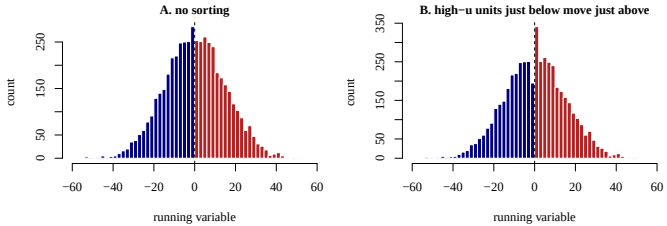
The Threat: Manipulation ★

Continuity fails if units can *precisely* control which side of the cutoff they land on, retaking a test until you pass, reporting income just under a threshold, a firm holding hiring just below a regulatory headcount.

- Imprecise control is fine. Lee (2008)'s insight is that if agents cannot control X to the last decimal, assignment near the cutoff is as good as random even though agents influence X.
- **Density test** (McCrary 2008; Cattaneo, Jansson, and Ma 2020): is the density of X continuous at c? A spike on the advantaged side is direct evidence of sorting.
- **Covariate smoothness**: run the RD with *pre-determined covariates* as outcomes. They should show no jump. This is the single most persuasive check, and it is the RD analogue of a balance table.
- **Donut hole**: drop observations immediately around c, where manipulation would concentrate, and see whether the estimate survives.

What Sorting Does

Suppose incumbents with a strong local machine can tip a close race (extra votes, a friendly recount, cheating). Simulated, with no true effect (figs/w9_sims.R):



- Rescued races move from just below 0 to just above: barely winning now often means "has a machine".
- The machine still works six years later, so part of the jump is machine capacity, not incumbency.
- Two traces: a hole below 0 and a spike above it (**density test**); winners who differed before the race (**balance test**).

The Density Test, and the Senate

The test. Estimate the density of X at c separately from each side and compare:

$$T = \frac{\hat{f}_+(c) - \hat{f}_-(c)}{\widehat{\text{se}}} \;\approx\; N(0,1) \quad \text{under no sorting.}$$

McCrary (2008) bins X and runs local linear regressions on the bin counts; Cattaneo, Jansson, and Ma (2020) fit a local polynomial to the empirical distribution function, with no bins (`rddensity`).

Senate: $T = -0.76$, $p = 0.45$; 49 elections lost by less than 2 points, 47 won by less than 2 (figs/w9_senate.R).

- A pass is weak evidence: sorting in *both* directions leaves the density smooth while changing who sits on each side.
- A failure is strong evidence, unless the spike has a mechanical cause such as rounding (next frames).

Covariate Balance on the Senate Data

Each covariate as the outcome; `rdrobust` at h_{CER}, bias-corrected jump, robust p:

covariate	mean below	jump	robust p
Democratic vote share, $t-6$	48.14	3.72	0.18
Democratic vote share, $t-12$	47.43	2.15	0.35
Democratic presidential vote share	44.97	−0.09	0.96
open seat	0.36	−0.14	0.09
state population (millions)	4.16	−0.25	0.77
House terms of the candidate	1.23	−0.24	0.66
Senate terms of the candidate	3.22	−0.10	0.90
midterm year	0.50	0.14	0.10
Democratic president	0.48	−0.07	0.47

- One of nine at $p < 0.10$; about one expected by chance. Watch the lagged vote: it moved the covariate-adjusted estimate (figs/w9_senate.R).
- If its jump were significant, seats Democrats barely won were already more Democratic before: strong seats tipping close races (the machine story), so part of the jump would be past strength, not winning.

Placebo Cutoffs and Donuts on the Senate Data

Placebo cutoffs (one side only):

cutoff	estimate	robust p
-20	6.62	0.16
-15	-6.89	0.23
-10	-23.22	0.005
-5	-0.18	0.72
5	3.32	0.67
10	2.92	0.11
15	-1.22	0.65
20	-0.25	0.89

Donut: drop $|X_i| < \delta$:

δ	dropped	estimate	robust 95%
0	0	7.41	[4.09, 10.92]
0.25	9	6.94	[3.32, 10.45]
0.5	23	7.09	[3.20, 10.83]
1	43	6.58	[2.19, 10.60]
2	96	4.70	[0.21, 8.96]

- One placebo of eight rejects at 1%: with eight tests that happens about 8% of the time by chance. Look at the plot near -10 before deciding.
- The donut shrinks the estimate as it removes the closest races, which carry the most weight; with $\delta = 2$ the fit extrapolates over a 4-point gap.

Heaping: a Cautionary Tale

The design (Almond, Doyle, Kowalski, and Williams 2010): newborns under 1,500 g are "very low birth weight" and get more care. One-year mortality just below is about 1 point lower (-0.0095, se 0.0048; 5.5% just above).

The problem (Barreca, Guldi, Lindo, and Waddell 2011): weights heap at round numbers, and heaped births are disproportionately poorer and less healthy. The heaps at 1,500 g and 53 oz (1,503 g) fall just *above* the cutoff, on the untreated side: they raise mortality there, so the care below looks better than it is.

Heap = pile up: many records land on round values because people round (a nurse writes 1,500 g for a baby of 1,497 g).

one-year mortality effect (85 g bandwidth)	estimate (se)
donut: drop multiples of 100 g	-0.0054 (0.0031)
donut: drop multiples of 100 g and of an ounce	0.0000 (0.0050)

IZA DP 5106 numbers; the original authors reply that the estimates fall but remain large (Almond et al. 2011).

When Sorting Is Present: Bounds (Gerard, Rokkanen, and Rothe 2020)

Suppose sorting only pushes units *up* past c. The density then has an excess just above c, and the excess is the movers:

$$\pi = \frac{f_+(c) - f_-(c)}{f_+(c)} = 1 - \frac{f_-(c)}{f_+(c)}$$

share of units just above c who moved there

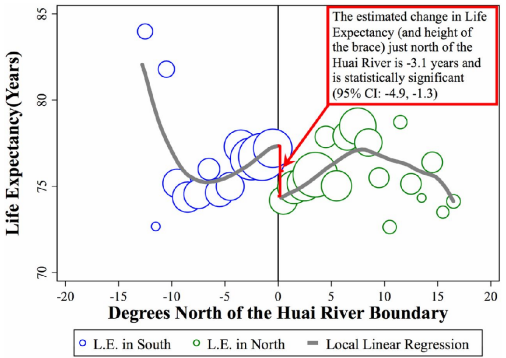
- The non-movers just above c are comparable to the units just below. We do not know *which* units above are movers, only how many.
- **Worst cases**: drop the top π share of outcomes above c, or the bottom π share. The two trimmed means bound $E[Y_i(1) \mid X_i = c, \text{non-mover}]$, and subtracting the mean just below bounds the effect.
- The trimming logic of bounds for sample attrition (Lee 2009). The bounds widen with π; a density test that fails by a little costs a little.

Other Checks and Common Errors

- **Placebo cutoffs**: estimate at fake thresholds away from c, using only data on one side so the real discontinuity is not picked up. Significant "effects" there mean your procedure finds jumps everywhere.
- **Bandwidth sensitivity**: plot $\hat{\tau}$ against h over a wide range with robust intervals. A result that exists only at one bandwidth is not a result.
- Do not use a high-order global polynomial (Gelman and Imbens 2019). The implied weights on distant observations are large, oscillating, and indefensible; estimates wander with the degree.
- Do not control for post-treatment covariates, and do not interact covariates with D unless you mean to estimate heterogeneity.
- Do not report only the conventional interval at an MSE-optimal bandwidth.

Why Not a Global Polynomial: the Huai River

Free coal heating only north of the Huai River. X: degrees north; Y: life expectancy.



Ebenstein et al. (2017), Fig. 3: disease-surveillance sites (circle size = population), local linear fit.

- **Chen et al. (2013)**: a cubic in latitude over all of China. North: 5.5 years lower (95% CI 0.8, 10.2).
- A global cubic lets sites 10–15 degrees away (left: 82–84 years) bend the curve, and the bend sets the gap at the river (Gelman and Imbens 2019).
- **Ebenstein et al. (2017)**: local linear near the river, newer data: 3.1 years lower (95% CI 1.3, 4.9).

Local Rand.

A Second Framework (Cattaneo, Frandsen, and Titiunik 2015)

Goal: treat a small window around c as an experiment, and analyse it as one.

Assumption: there is a window $W_0 = [c - w,\ c + w]$ in which

1. which units fall above c is as if decided by a coin flip, and
2. $Y_i(d)$ does not depend on X_i inside the window.

- Then the difference in means inside W_0 estimates the effect, and a **Fisher randomization test** gives exact p-values: reshuffle who is above c among the window's units, recompute, compare.
- **Continuity** extrapolates two curves to one point; **local randomization** assumes a whole window is an experiment. Condition 2 is stronger, and it is why w must be small (Sekhon and Titiunik 2017).
- Natural when X is discrete with few values near c (ranks, ages in years), where curve fitting has little to work with.

Choosing the Window, Then Testing

Window: in nested windows, test balance on pre-determined covariates; take the largest window in which the smallest p-value stays above 0.15 (`rdlocrand::rdwinselect`). Senate, nine covariates (figs/w9_senate.R):

window	smallest balance p	below	above
± 0.500	0.225	9	14
± 0.625	0.208	13	17
± 0.750	0.065	15	22
± 1.000	0.043	17	25

Test in ± 0.625: mean vote at $t + 6$ is 42.74 below and 52.36 above, a difference of 9.62. Of 5,000 reshufflings of the 17 "above" labels among the 30 races, 0.12% give a difference this large: $p = 0.0012$ (permutation test).

Larger than the continuity estimate (7.41), from 30 races instead of 683. Cattaneo, Frandsen, and Titiunik (2015) use ± 0.75 (9.69 here).

In the Literature: Bol, de Vaan, and van de Rijt (2018)

The question. Merton's Matthew effect: does early success cause later success, or do repeated winners just have more ability? Cumulative advantage is a cornerstone claim of stratification research and hard to separate from selection.

The data. The Dutch science foundation's early-career "Veni" grant (€250,000): review scores and decisions for recent PhDs, followed for up to 16 years.

The design. A hard threshold: every winner scored above every non-winner. Running variable: the applicant's rank relative to the funding line in each competition; compare the two ranks just above (+1, +2) with the two just below (−1, −2), with exact tests. A sharp RD with a discrete running variable, analysed as a local experiment.

Outcomes. Later grants (the mid-career "Vidi" and three others), applications for them, publications and citations, a full professorship.

Bol, de Vaan, and van de Rijt: Checking the Design

No sorting	Applicants cannot place their score; ranks are set by panels. A density test is uninformative: ranks are evenly spaced by construction.
Continuity	Publications, citations, and H-index *before* the competition rise smoothly with rank, with no jump at the line.
Placebo cutoffs	No difference in later outcomes between any other pair of neighbouring ranks.
Specification	A parametric RD (supplement) gives the same size and significance.

- Only rank is used, not the score; the comparison of $\pm 1, 2$ is the local-randomization window, chosen before looking.
- Publications, citations, and H-index *after* the competition also show no jump at the line: the later funding gap is not explained by grant-enabled output.

Bol, de Vaan, and van de Rijt: Main Results

Ranks $+1/+2$ against $-1/-2$:

outcome	just below	just above	test
won a mid-career grant	10%	26%	$P < 0.001$
applied for a mid-career grant	40%	59%	$P < 0.001$
later funding, 4 years on		+€40,000	$P < 0.01$
later funding, 8 years on		+€180,000	$P < 0.01$
full professor in 2018	19%	28%	$P < 0.05$

- The near-winners got more than twice the later money of near-losers, excluding the Veni itself; this is 40% of the gap between best and worst.

Fuzzy

When the Rule Is Not Followed ★

Often crossing the cutoff changes the *probability* of treatment rather than determining it: eligibility is not take-up.

$$\lim_{x \downarrow c} \Pr(D_i = 1 \mid X_i = x) - \lim_{x \uparrow c} \Pr(D_i = 1 \mid X_i = x) \;\in\; (0, 1).$$

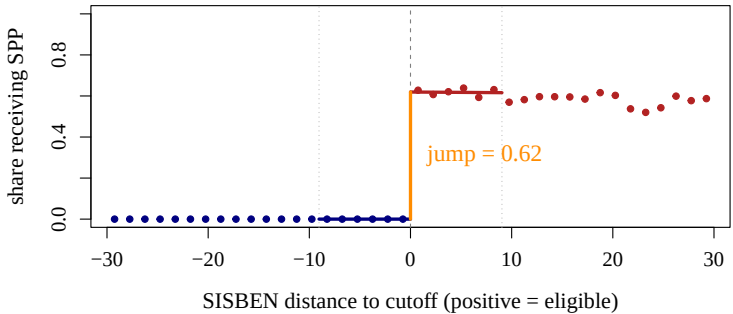
The fuzzy RD estimator is a ratio of two sharp RD estimates:

$$\tau_{FRD} = \frac{\text{jump in } Y \text{ at } c}{\text{jump in } D \text{ at } c} = \frac{\lim_{x \downarrow c} E[Y \mid x] - \lim_{x \uparrow c} E[Y \mid x]}{\lim_{x \downarrow c} E[D \mid x] - \lim_{x \uparrow c} E[D \mid x]}.$$

- This is exactly the Wald estimator of Week 7, with $\mathbb{1}\{X \geq c\}$ as the instrument, estimated locally.
- The numerator alone is the **intention-to-treat** effect of crossing the cutoff, and it is worth reporting on its own, it needs no first-stage assumption.

The Jump in D: Ser Pilo Paga

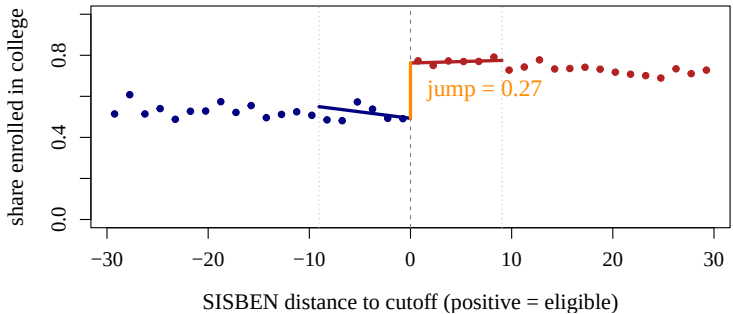
Londoño-Vélez, Rodríguez, and Sánchez (2020), Lab 9: college tuition for poor, high-scoring students, eligible below a wealth cutoff. D_i = received it.



- Below the cutoff no one receives it; above, 62% do: the jump in D is 0.62, not 1. The denominator of the ratio.
- Local linear, triangular kernel, $h = 9.04$ (dotted): `rdrobust`'s h for the ratio.

The Jump in Y: Ser Pilo Paga

Same students, same h; Y_i = enrolled in college right after the exam.



- The jump in Y is 0.27: eligibility raises enrollment by 27 points (intention to treat). The numerator of the ratio.
- $\hat{\tau}_{FRD} = 0.269/0.619 = 0.435$, the `rdrobust` fuzzy estimate: receiving SPP raises enrollment by 43 points for compliers at the cutoff.

What the Ratio Estimates: Four Types at the Cutoff

Write $D_i(1)$ for the treatment unit i would take just above c and $D_i(0)$ for the treatment just below. As in Week 7, every unit is one of four types:

Type	$D_i(0)$, just below	$D_i(1)$, just above	
Compliers	0	1	treated only if above c
Always-takers	1	1	treated on both sides
Never-takers	0	0	untreated on both sides
Defiers	1	0	treated only if below c

Three assumptions, each the RD version of a Week 7 one:

- **Continuity** (replaces independence): the share of each type and each type's mean $Y_i(1)$ and $Y_i(0)$ do not jump at c: the same mix of people just below and just above.
- **Exclusion**: crossing c changes Y only through D.
- **Monotonicity**: no defiers, i.e., no one is treated because they fell below the cutoff.

What the Ratio Estimates: the Numerator

Split the jump in Y by type. By continuity, both limits use the same shares and means, those at $X_i = c$, written $\Pr(\mathrm{C})$, $E[Y_i(1) \mid \mathrm{C}]$, and so on:

$$\begin{aligned}
&\lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x] \\
&\quad = \Pr(\mathrm{AT}) \underbrace{\left(E[Y_i(1) \mid \mathrm{AT}] - E[Y_i(1) \mid \mathrm{AT}]\right)}_{=0} \\
&\qquad + \Pr(\mathrm{NT}) \underbrace{\left(E[Y_i(0) \mid \mathrm{NT}] - E[Y_i(0) \mid \mathrm{NT}]\right)}_{=0} \\
&\qquad + \Pr(\mathrm{C}) \ \left(E[Y_i(1) \mid \mathrm{C}] - E[Y_i(0) \mid \mathrm{C}]\right) \\
&\qquad - \Pr(\mathrm{D}) \ \left(E[Y_i(1) \mid \mathrm{D}] - E[Y_i(0) \mid \mathrm{D}]\right).
\end{aligned}$$

Always- and never-takers take the same treatment on both sides, so by exclusion they contribute zero. Monotonicity removes the last line.

What the Ratio Estimates: the Denominator

Just above c the treated are compliers and always-takers; just below, always-takers and defiers:

$$\begin{aligned} \text{jump in } D &= \big[\Pr(\text{C}) + \Pr(\text{AT})\big] - \big[\Pr(\text{AT}) + \Pr(\text{D})\big] && \text{by type} \\ &= \Pr(\text{C}) - \Pr(\text{D}) \\ &= \Pr(\text{C}) && \text{no defiers} \end{aligned}$$

Dividing:

The fuzzy RD estimand (Hahn, Todd, and van der Klaauw 2001)

$$\tau_{FRD} = \frac{\text{jump in } Y}{\text{jump in } D} = E\big[\, Y_i(1) - Y_i(0) \;\big|\; i \text{ is a complier},\ X_i = c \,\big].$$

The Week 7 LATE theorem with "at the cutoff" added: the effect for compliers whose running variable is at c.

Estimation, Route 1: Two Sharp RDs and Their Ratio

Keep units with $|X_i - c| \le h$, weight them by the kernel, let $Z_i = \mathbb{1}\{X_i \ge c\}$, and run the *same* sharp RD twice:

$$D_i = \alpha_D + \beta_D Z_i + \gamma_D (X_i - c) + \eta_D Z_i (X_i - c) + v_i \qquad \text{sharp RD on } D\text{: jump } \hat{\beta}_D$$

$$Y_i = \alpha_Y + \beta_Y Z_i + \gamma_Y (X_i - c) + \eta_Y Z_i (X_i - c) + e_i \qquad \text{sharp RD on } Y\text{: jump } \hat{\beta}_Y$$

$$\hat{\tau} = \hat{\beta}_Y / \hat{\beta}_D \qquad \text{Wald ratio, Week 7}$$

- This is IV with Z_i as the instrument: $\hat{\beta}_D$ is the first stage, $\hat{\beta}_Y$ the reduced form.
- Both regressions must use the same h, kernel, and polynomial; otherwise the ratio is not an IV estimate of anything.
- Ser Pilo Paga, $h = 9.04$: $0.269/0.619 = 0.435$.

Estimation, Route 2: Local 2SLS

Same window, same kernel, same instrument Z_i. As in Week 7:

$$\text{Stage 1:} \quad D_i = \alpha_D + \beta_D Z_i + \gamma_D(X_i - c) + \eta_D Z_i(X_i - c) + v_i \quad \Longrightarrow \quad \hat{D}_i$$

$$\text{Stage 2:} \quad Y_i = \alpha + \tau\, \hat{D}_i + \gamma(X_i - c) + \eta\, Z_i(X_i - c) + u_i$$

- Stage 1 is Route 1's sharp RD on D. The two slope terms are controls and enter *both* stages; Z_i is the only excluded instrument.
- One instrument, one endogenous D: 2SLS equals the Route 1 ratio exactly. Ser Pilo Paga: `ivreg` gives 0.435.
- Do not take the standard error from the second stage by hand (Week 7): use `rdrobust(y, x, fuzzy = d)`. Report $\hat{\beta}_D$ in the main table.
- `rdrobust` picks one MSE-optimal h for the ratio $\hat{\tau}$ and uses the *same* h in both stages, so ratio = 2SLS still holds.

Fuzzy RD Inherits Everything From IV ★

- **Exclusion**: crossing the cutoff must affect Y only through D. If the threshold also triggers something else, a different school, a different caseworker, a label, exclusion fails.
- **Monotonicity**: no one is treated *because* they fell below the cutoff.
- **The estimand is doubly local**: the effect for compliers at the cutoff. Not the ATE, not the ATT, and not the effect for everyone near the cutoff.
- **Weak first stage** is a real risk: if take-up jumps by only a few percentage points, you are dividing by a small noisy number and all the Week 7 pathologies return. Check strength with the same first-stage F as in Week 7: the jump in D and its F statistic.

Vietnam

In the Literature: Dell and Querubín (2018)

The question. Does force from above win hearts and minds? The effect of U.S. bombing on insurgency, local government, and civic life in South Vietnam, and of the Army's firepower doctrine against the Marines' "hearts and minds".

The data. The Hamlet Evaluation System (HES), 1969–1973: 169 monthly and quarterly questions on security, politics, and the economy, answered by U.S. advisers for nearly every hamlet. Outcomes: indices built from HES answers, armed-forces records of operations and attacks, and public opinion surveys.

Two designs.

1. **Bombing**: a fuzzy RD at the rounding thresholds of an algorithm that scored hamlet security; air strikes were targeted on the rounded score.
2. **Doctrine**: a boundary RD at the line between the Marines' I Corps and the Army's II Corps.

Numbers in this section are from NBER Working Paper 22395 (2016); the QJE version is *QJE* 133(2): 701–764.

The Score Nobody Saw

- The HES questions feed 19 submodels. Each starts from a flat prior over five security classes, A (5) to E (1), and updates by Bayes' rule using a matrix of conditional probabilities.
- The posterior mean, a continuous number between 1 and 5, is rounded: 3.49 becomes C, 3.51 becomes B. Rounded submodel scores are combined into one letter grade per hamlet.
- Air strikes were planned weekly, with the hamlet grade among the inputs. Just below a threshold, a hamlet is graded one letter worse than a near-identical one just above.
- The continuous scores were never printed; the conditional probabilities were classified. The authors rebuilt them from archived tapes and reproduce every rounded score.

Running variable: distance from the continuous score to the nearest of four thresholds (A/B, B/C, C/D, D/E); **instrument**: being below it. Hamlets near each threshold: 35%, 46%, 16%, 3%. **Pooling four thresholds**: the pooled effect is a weighted average of the threshold-specific effects, weights $\propto$ hamlets at each threshold, so it is mostly about B/C hamlets (Cattaneo, Keele, Titiunik, and Vazquez-Bare 2016).

Dell and Querubín: the Estimating Equation

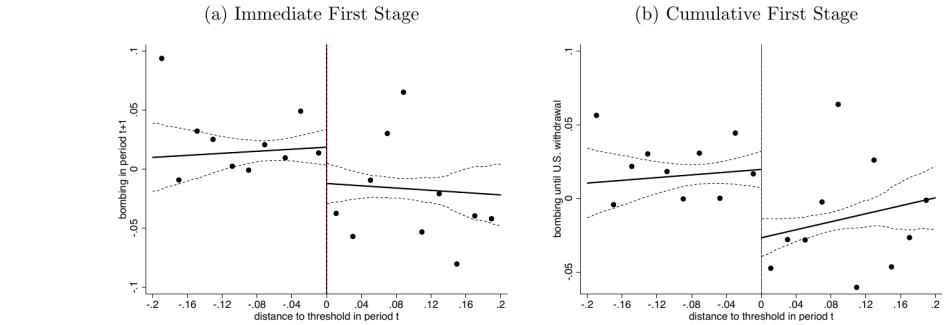
For hamlet h first near a threshold in quarter t (each hamlet once): $Z_{ht} = 1$ if just *below* the threshold (instrument), D_{ht} = bombing (treatment), $Y_{h,t+n}$ = outcome n quarters later.

$$\text{Stage 1:} \quad D_{ht} = \pi Z_{ht} + \sum_{d=1}^{4} \Big(\delta_d G_{htd} + G_{htd}\, f_d(\text{dist}_{ht}) + G_{htd}\, f_d(\text{dist}_{ht})\, Z_{ht} \Big) + \alpha_t + X'_{ht}\beta + v_{ht}$$

$$\text{Stage 2:} \quad Y_{h,t+n} = \tau \hat{D}_{ht} + (\text{the same threshold terms, } \alpha_t,\ X'_{ht}\beta) + \epsilon_{ht}$$

- $G_{htd} = 1$ if threshold d is the nearest: its own intercept and local linear slopes on each side of each threshold.
- α_t: quarter fixed effects; X_{ht}: the answers that entered the score, for precision. IK bandwidth; s.e. clustered by village.
- $\hat{\pi}$ is the first stage; Y on Z_{ht} with the same terms is the reduced form. $\hat{\tau}$ is a LATE for hamlets bombed *because* they were rounded down.

Dell and Querubín: the First Stage



Their Figure 3(a)–(b). Horizontal: distance to the threshold; below the threshold on the left. Vertical: bombing, residualized.

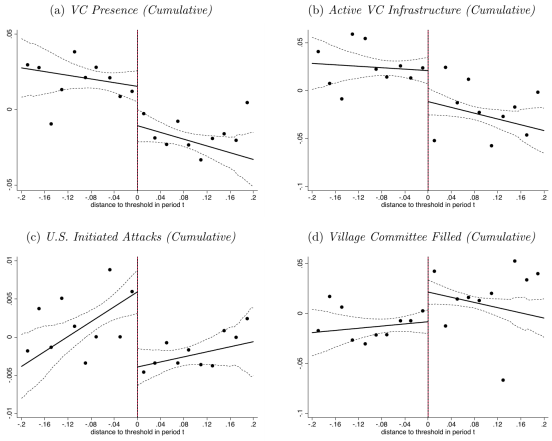
- Being just below raises the share of months with air or artillery strikes near the hamlet next quarter by 0.054 (0.014), from a mean of 0.28; $F = 14.9$. Through U.S. withdrawal: 0.044 (0.013), mean 0.26, $F = 11.5$.

Dell and Querubín: Checking the Design

No sorting	Scores were computed centrally and never printed; field staff did not know the probabilities. McCrary test: no jump in the density.
Continuity	Pre-period outcomes balanced (a few of many differ, e.g. South Vietnamese deaths). Bombing in the current and previous quarter does not jump: 0.011 (0.011), −0.001 (0.018).
Placebo year	In 1969 the score was computed but not used: no first stage, −0.016 (0.019).
Exclusion	No jump in other allocations (ground troops, U.S. operations, naval attacks, local forces): friendly forces nearby 0.016 (0.018).

- The 1969 placebo is the strongest check: same algorithm, same thresholds, no use of the score, no jump. Whatever moves bombing in 1970 is the score's use, not hamlet differences at the threshold.
- Exclusion is argued by measuring the other channels a lower grade could trigger.

Dell and Querubín: Reduced Forms



Their Figure 5(a)–(d), cumulative outcomes through withdrawal. Left of zero: rounded down, bombed more.

Dell and Querubín: Main Results

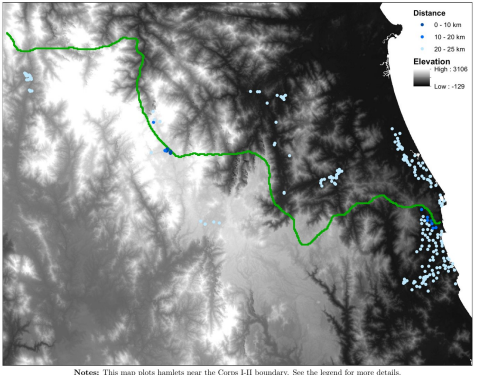
IV estimates, outcomes cumulative through U.S. withdrawal (their Tables 4, 6, 7):

outcome	IV (se)	mean	at mean bombing
security index	−0.642 (0.246)	0.68	−17 pts
armed Viet Cong present	0.571 (0.222)	0.19	+15 pts
Viet Cong infrastructure active	0.978 (0.384)	0.25	+25 pts
village committee filled	−0.798 (0.380)	0.84	−21 pts
local government collects taxes	−0.944 (0.443)	0.70	−25 pts
civic society index	−0.523 (0.248)	0.69	−14 pts

- **Reading**: a coefficient is the effect of going from no bombing to bombing in every month; the last column multiplies by mean bombing, 0.26 (e.g. $0.571 \times 0.26 = 0.15$).
- Bombing strengthened the insurgency it aimed at and weakened local government and non-communist civic life.

Dell and Querubín: a Boundary Design

Figure 7: Corps Region Boundary



Notes: This map plots hamlets near the Corps I-II boundary. See the legend for more details.

Their Figure 7: the Corps I/II boundary; hamlets within 25 km.

The question: does *how* a war is fought matter, not only how much?

Two doctrines, one line.

- Corps I (north), the **Marines**: "Civic Action" (development aid) and "Combined Action" (small units living in villages with local forces).
- Corps II (south), the **Army**: overwhelming firepower, search and destroy, body counts.

Treatment: being on the Marine side. **Running variable**: the hamlet's location. **Outcomes**: HES security, insurgency, and public goods, averaged up to the Marines' withdrawal (April 1971).

Geography as a Running Variable

The running variable is location (lat, lon); the cutoff is a *line*, not a point (Keele and Titiunik 2015).

- **Assumption**: everything else that affects Y changes smoothly across the line, so hamlets just north and just south are comparable. It is the continuity assumption of the sharp RD, in two dimensions.
- **Estimand**: a jump $\tau(b)$ at every point b on the line. One number averages them, weighted by where units live along the line.
- **Checks**: the sharp-RD ones, on a map: balance in terrain and in pre-period outcomes; several bandwidths (km); drop units right at the line if treatment can spill across it.
- Compound treatment is the rule: borders separate taxes, schools, police, and history at once. Name the bundle.

Dell and Querubín: the Specification

For hamlet h on boundary segment s, within 25 km of the line:

$$Y_{hs} = \alpha_0 + \tau\,\text{Marines}_{hs} + f(\text{lat}_{hs}, \text{lon}_{hs}) + G'_{hs}\beta + \alpha_s + \epsilon_{hs}$$

f: local linear in latitude and longitude; G: geographic controls; α_s: two boundary segments; s.e. clustered by village; 302 hamlets, 64 villages.

- τ is the jump at the line, averaged along it: Marines minus Army for neighbouring hamlets.
- Identifying assumption: "all other factors change smoothly at the boundary". The authors do not argue the line was drawn arbitrarily; balance has to carry the argument.
- Bombing, the first design's treatment, is 11 points lower on the Marine side, not significant (s.e. 0.10).

Dell and Querubín: Balance at the Boundary

The same regression with a pre-period or fixed characteristic as Y (their Table 9):

characteristic	Marines side (se)	mean
VC attacks, 1964 to May 1965	0.020 (0.047)	
elevation (m)	−78 (108)	
slope	−1.55 (1.50)	
market nearby, 1929 map	−0.032 (0.040)	
military post nearby, 1929 map	0.300 (0.160)	0.05
road km, 1929 map	2.253 (0.816)	1.63
paved road km, 1929 map	−0.109 (0.710)	

Pre-war violence and terrain are balanced; colonial posts (10% level) and unpaved roads are not.

Dell and Querubín: Boundary Results

Effect of being on the Marine side (their Table 10):

outcome	Marines side (se)	mean
armed Viet Cong present	−0.560 (0.075)	0.32
VC-initiated attacks on the hamlet	−0.482 (0.061)	0.33
VC attacks on troops	−0.191 (0.071)	0.18
health index	0.562 (0.198)	0.36
education index	0.243 (0.096)	0.25
security index	0.095 (0.110)	0.35

- The Marine side had far less insurgent presence and more public goods.
- Not just less bombing: bombing did not differ significantly at the line.
- Robust to a quadratic and to 50 km (education loses significance), and to dropping hamlets within 10 km of the line (spillovers).

Dell and Querubín: What the Boundary Cannot Separate

- **Compound treatment.** More than doctrine changes at the line: the service, its soldiers, officer rotation (six months in combat for Army officers, up to a year for Marines). The authors grant these channels cannot be ruled out.
- **Balance is imperfect**: colonial military posts 0.300 (0.160); total roads 2.253 (0.816), more on the Marine side.
- **Few clusters**: 64 villages; attitudes come from 13 surveyed hamlets near the line, so that table is OLS within 100 km, not RD.
- **Comparison of the two designs**: the bombing RD has a single, measured treatment and a placebo year; the boundary RD compares two bundles of everything an army does.

The headline survives both: the firepower approach produced worse outcomes, on security and on the institutions it meant to build.

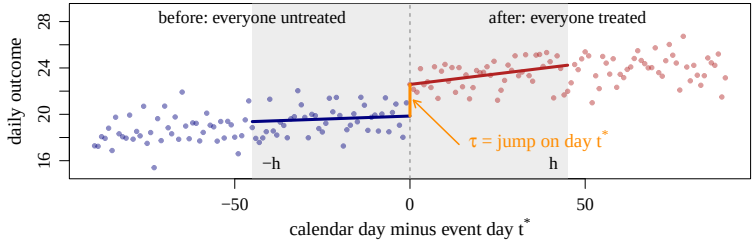
Extensions

RD in Time: the Setup

Goal: the effect of an event that starts on one date t^* for everyone (Hausman and Rapson 2018). The running variable is the date t_i of observation i; keep observations with $|t_i - t^*| \leq h$ days:

$$Y_i = \alpha + \tau \mathbb{1}\{t_i \geq t^*\} + \beta (t_i - t^*) + \gamma \mathbb{1}\{t_i \geq t^*\}(t_i - t^*) + \epsilon_i$$

τ is the jump on the day of the event: the sharp RD, with days for X.



Hypothetical data. Unlike turning 21 (each person's own birthday), t^* is one calendar date for all: anything else that happened that day also jumps.

In the Literature: Legewie (2016)

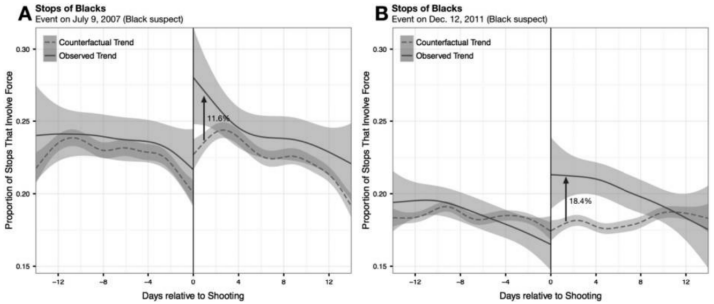
The question. Do salient events trigger periods of more racially biased policing? Two NYPD officers were shot by Black suspects (July 9, 2007; December 12, 2011).

The data. About 3.9 million NYPD pedestrian stops, 2006–2012. Outcome: any physical force in the stop (16% of stops of whites, 22% of Blacks, 24% of Hispanics).

The design. Time is the running variable; the shooting is the cutoff. The main analysis compares stops in the three days after each shooting with matched stops (same precinct, race, place, time of day, suspect characteristics) from the year before; an RD in time with local linear fits is the robustness check.

three days after, log-odds of force (se)	White	Black	Hispanic
December 2011	−0.30 (0.16)	0.20 (0.05)	−0.10 (0.08)
July 2007	−0.01 (0.14)	0.16 (0.06)	0.02 (0.08)

Legewie: Results and Checks

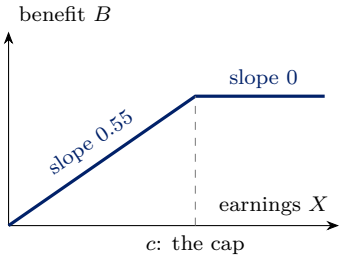


His Figure 3: force in stops of Black suspects, observed against counterfactual trend.

- Force against Blacks up 16% (21.9% to 25.4%) in 2011, 13% in 2007; no change for whites or Hispanics; it fades in about 3.5 and 10 days.
- **Placebo events**: shootings by white or Hispanic suspects show no effect. Among 800 random sets of dates, 1.3% match the Black-suspect pattern.
- **RD in time**: estimates across bandwidths stay positive; very short windows shrink the 2007 effect, as news needs time to spread.

Regression Kink: the Rule Bends Instead of Jumping

Austrian unemployment benefits (Card, Lee, Pei, and Weber 2015): the daily benefit is 55% of last year's net daily earnings, up to a maximum.



- At the cap, no one's benefit *jumps*: a sharp RD finds nothing to compare.
- The *slope* of benefits in earnings drops from 0.55 to 0.
- If higher benefits lengthen unemployment, unemployment duration plotted against X must bend at the same c.

Effect of one euro of benefit = the change in the slope of Y divided by the change in the slope of B.

Hypothetical: duration rises 0.11 days per euro of earnings below the cap and 0.01 above. Effect $= (0.01 - 0.11)/(0 - 0.55) = 0.18$ days per euro of benefit.

Regression Kink: the Estimator

Fit a line on each side of c, as in the sharp RD, and keep the change in slope:

$$B_i = a_B + \beta_B(X_i - c) + \kappa_B\,(X_i - c)\,\mathbb{1}\{X_i \geq c\} + u_i$$
$$Y_i = a_Y + \beta_Y(X_i - c) + \kappa_Y\,(X_i - c)\,\mathbb{1}\{X_i \geq c\} + \epsilon_i, \qquad \hat{\tau}_{RK} = \hat{\kappa}_Y / \hat{\kappa}_B$$

No jump term: B and Y are continuous at c; only their slopes change.

- **Assumption**: nothing else bends at c; the density of X has no kink there (testable).
- It is the fuzzy-RD ratio with slopes for levels: κ_B is the first stage.

Regression Kink: What Card, Lee, Pei, and Weber Find

- **Data**: about 275,000 Austrian UI claims, 2001–2012, by job losers with a year of work in the past two years. Y: days of registered unemployment (capped at one year).
- **Fuzzy kink**: supplements for dependents are not observed, so the observed benefit does not follow the formula exactly; κ_B is estimated.
- **Result**: the elasticity of unemployment duration with respect to the benefit is 1.4 to 1.9 across estimators at larger bandwidths: a 10% higher benefit, 14–19% longer unemployment.
- **The catch**: slopes are estimated far less precisely than levels. At small bandwidths the estimates are unstable, and only one of three robust intervals excludes zero. Kinks need much more data than jumps.

Repeated Cutoffs: the Problem

School bonds in California (Cellini, Ferreira, and Rothstein 2010): a district borrows for buildings only if voters approve (two-thirds; 55% from 2001). Does a new bond raise local house prices?

A close vote is a sharp RD at the vote threshold. But a district that *barely loses* can propose again next year, and many do:

	year t	year $t+1$	year $t+2$
barely won at t	bond	bond, 1 year old	bond, 2 years old
barely lost at t	no bond	some pass a bond	more have a bond

- By $t+2$ we compare "a 2-year-old bond" with "maybe a newer bond": the gap understates what a bond does.
- The same problem as non-compliance in a fuzzy RD, except that the losers' take-up grows with time.

Repeated Cutoffs: Two Effects

Three RD estimates, all at the threshold of the vote in year t:

symbol	what it is	estimated by
ITT_k	total effect of barely winning at t on Y_{t+k}, later bonds included	RD of Y_{t+k}
π_j	jump in the chance of passing a bond in year $t+j$; $\pi_0 = 1$, $\pi_j < 0$ if losers catch up	RD of "passed at $t+j$"
θ_k	**direct** effect of one bond, k years after it passes	**the target**

- ITT_k answers "what does winning this vote do?" (retries included). θ_k answers "what does a bond do?".
- Cellini et al. call θ_k the TOT and ITT_k the ITT.

Repeated Cutoffs: Peeling the ITT, Year by Year

Each bond induced in year $t+j$ has acted for $k-j$ years by $t+k$:

$$\mathrm{ITT}_0 = \theta_0 \qquad \text{no time to retry}$$

$$\mathrm{ITT}_1 = \theta_1 + \pi_1\theta_0 \qquad \text{winners' bond is 1 year old; losers' new bonds are 0}$$

$$\mathrm{ITT}_2 = \theta_2 + \pi_1\theta_1 + \pi_2\theta_0 \qquad \text{bonds passed at } t+1 \text{ are now 1 year old}$$

$$\theta_k = \mathrm{ITT}_k - \textstyle\sum_{j=1}^{k} \pi_j\,\theta_{k-j} \qquad \text{solve from } \theta_0 \text{ upward}$$

hypothetical	$k=0$	$k=1$	$k=2$
ITT_k (% house prices)	10	12	11
π_k	1	-0.4	-0.1
θ_k	10	$12 + 0.4(10) = 16$	$11 + 0.4(16) + 0.1(10) = 18.4$

The ITT falls after year 1 while the bond's own effect keeps rising: losers are catching up.

Repeated Cutoffs: Assumptions and Results

Assumptions of the recursion:

- Each close vote is a valid RD (continuity at its threshold).
- θ_k is the same for a bond passed at the first try and one passed after a loss, whatever the district's history; effects of several bonds add up. Hsu and Shen (2024) drop the first part, allow heterogeneous effects, and find much larger long-run direct effects on capital spending.

Results:

- Cellini et al.: prices rise soon after a bond passes and stay up; homebuyers pay \$1.50 or more per \$1 of school capital spending.
- Baron (2022), Wisconsin, separate votes on operating and capital budgets: operating money raises test scores and college enrollment; buildings do little.

Practice

Reading and Reporting an RD ★

1. **The picture first.** Binned means of Y against X, with the fitted local polynomials and the cutoff. If the jump is not visible, be sceptical of the number.
2. The bandwidth, how it was chosen, and the effective n on each side.
3. Robust bias-corrected confidence intervals.
4. The density test and the covariate-smoothness checks.
5. A bandwidth-sensitivity plot and placebo cutoffs.
6. For fuzzy designs, the size of the jump in treatment take-up, and the intention-to-treat effect alongside.
7. An explicit statement that the estimand is local to the cutoff.

Looking Ahead

- RD buys extraordinary credibility at the cost of a very local estimand. The identifying variation is visible in a scatterplot, and the main threats (sorting, bandwidth dependence) are checkable.
- Note the family resemblance: sharp RD is a local experiment; fuzzy RD is local IV. The tools recombine rather than multiply.
- **Next week**: designs that use variation over *time* rather than across a threshold.
- Note again how the estimand was chosen by the design rather than by us. Saying so plainly is part of reporting an RD honestly.

Problem sets

Problem Set 3 is assigned today and is due Tuesday, November 3. It covers matching (Week 6), instrumental variables (Week 7), and regression discontinuity (Week 9).

References

- Almond, Douglas, Joseph J. Doyle, Amanda E. Kowalski, and Heidi Williams. 2010. "Estimating Marginal Returns to Medical Care: Evidence from At-Risk Newborns." *Quarterly Journal of Economics* 125(2): 591–634.
- Almond, Douglas, Joseph J. Doyle, Amanda E. Kowalski, and Heidi Williams. 2011. "The Role of Hospital Heterogeneity in Measuring Marginal Returns to Medical Care: A Reply to Barreca, Guldi, Lindo, and Waddell." *Quarterly Journal of Economics* 126(4): 2125–2131.
- Armstrong, Timothy B. and Michal Kolesár. 2018. "Optimal Inference in a Class of Regression Models." *Econometrica* 86(2): 655–683.
- Armstrong, Timothy B. and Michal Kolesár. 2020. "Simple and Honest Confidence Intervals in Nonparametric Regression." *Quantitative Economics* 11(1): 1–39.
- Barreca, Alan I., Melanie Guldi, Jason M. Lindo, and Glen R. Waddell. 2011. "Saving Babies? Revisiting the Effect of Very Low Birth Weight Classification." *Quarterly Journal of Economics* 126(4): 2117–2123.
- Baron, E. Jason. 2022. "School Spending and Student Outcomes: Evidence from Revenue Limit Elections in Wisconsin." *American Economic Journal: Economic Policy* 14(1): 1–39.

References (cont.)

- Bol, Thijs, Mathijs de Vaan, and Arnout van de Rijt. 2018. "The Matthew Effect in Science Funding." *Proceedings of the National Academy of Sciences* 115(19): 4887–4890.
- Calonico, Sebastian, Matias D. Cattaneo, and Max H. Farrell. 2018. "On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference." *Journal of the American Statistical Association* 113(522): 767–779.
- Calonico, Sebastian, Matias D. Cattaneo, and Max H. Farrell. 2020. "Optimal Bandwidth Choice for Robust Bias-Corrected Inference in Regression Discontinuity Designs." *Econometrics Journal* 23(2): 192–210.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica* 82(6): 2295–2326.
- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocío Titiunik. 2019. "Regression Discontinuity Designs Using Covariates." *Review of Economics and Statistics* 101(3): 442–451.
- Carpenter, Christopher and Carlos Dobkin. 2009. "The Effect of Alcohol Consumption on Mortality: Regression Discontinuity Evidence from the Minimum Drinking Age." *American Economic Journal: Applied Economics* 1(1): 164–182.

References (cont.)

- Card, David, David S. Lee, Zhuan Pei, and Andrea Weber. 2015. "Inference on Causal Effects in a Generalized Regression Kink Design." *Econometrica* 83(6): 2453–2483.
- Cattaneo, Matias D., Brigham R. Frandsen, and Rocío Titiunik. 2015. "Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate." *Journal of Causal Inference* 3(1): 1–24.
- Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik. 2020. *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge Elements. Cambridge University Press.
- Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik. 2024. *A Practical Introduction to Regression Discontinuity Designs: Extensions*. Cambridge Elements. Cambridge University Press.
- Cattaneo, Matias D., Michael Jansson, and Xinwei Ma. 2020. "Simple Local Polynomial Density Estimators." *Journal of the American Statistical Association* 115(531): 1449–1455.

References (cont.)

- Cattaneo, Matias D., Luke Keele, Rocío Titiunik, and Gonzalo Vazquez-Bare. 2016. "Interpreting Regression Discontinuity Designs with Multiple Cutoffs." *Journal of Politics* 78(4): 1229–1248.
- Cattaneo, Matias D., Rocío Titiunik, and Gonzalo Vazquez-Bare. 2019. "Power Calculations for Regression-Discontinuity Designs." *Stata Journal* 19(1): 210–245.
- Cellini, Stephanie Riegg, Fernando Ferreira, and Jesse Rothstein. 2010. "The Value of School Facility Investments: Evidence from a Dynamic Regression Discontinuity Design." *Quarterly Journal of Economics* 125(1): 215–261.
- Caughey, Devin and Jasjeet S. Sekhon. 2011. "Elections and the Regression Discontinuity Design: Lessons from Close U.S. House Races, 1942–2008." *Political Analysis* 19(4): 385–408.
- Chen, Yuyu, Avraham Ebenstein, Michael Greenstone, and Hongbin Li. 2013. "Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *PNAS* 110(32): 12936–12941.

References (cont.)

- Dell, Melissa and Pablo Querubín. 2018. "Nation Building Through Foreign Intervention: Evidence from Discontinuities in Military Strategies." *Quarterly Journal of Economics* 133(2): 701–764.
- Ebenstein, Avraham, Maoyong Fan, Michael Greenstone, Guojun He, and Maigeng Zhou. 2017. "New Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy." *PNAS* 114(39): 10384–10389.
- Eggers, Andrew C., Anthony Fowler, Jens Hainmueller, Andrew B. Hall, and James M. Snyder. 2015. "On the Validity of the Regression Discontinuity Design for Estimating Electoral Effects: New Evidence from Over 40,000 Close Races." *American Journal of Political Science* 59(1): 259–274.
- Gelman, Andrew and Guido Imbens. 2019. "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs." *Journal of Business & Economic Statistics* 37(3): 447–456.
- Gerard, François, Miikka Rokkanen, and Christoph Rothe. 2020. "Bounds on Treatment Effects in Regression Discontinuity Designs with a Manipulated Running Variable." *Quantitative Economics* 11(3): 839–870.

References (cont.)

- Goldberger, Arthur S. 1972. "Selection Bias in Evaluating Treatment Effects: Some Formal Illustrations." Discussion Paper 123-72, Institute for Research on Poverty, University of Wisconsin.
- Hahn, Jinyong, Petra Todd, and Wilbert van der Klaauw. 2001. "Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design." *Econometrica* 69(1): 201–209.
- Hausman, Catherine and David S. Rapson. 2018. "Regression Discontinuity in Time: Considerations for Empirical Applications." *Annual Review of Resource Economics* 10: 533–552.
- Hsu, Yu-Chin and Shu Shen. 2024. "Dynamic Regression Discontinuity under Treatment Effect Heterogeneity." *Quantitative Economics* 15(4): 1035–1064.
- Imbens, Guido W. and Thomas Lemieux. 2008. "Regression Discontinuity Designs: A Guide to Practice." *Journal of Econometrics* 142(2): 615–635.
- Imbens, Guido W. and Karthik Kalyanaraman. 2012. "Optimal Bandwidth Choice for the Regression Discontinuity Estimator." *Review of Economic Studies* 79(3): 933–959.

References (cont.)

- Keele, Luke J. and Rocío Titiunik. 2015. "Geographic Boundaries as Regression Discontinuities." *Political Analysis* 23(1): 127–155.
- Kolesár, Michal and Christoph Rothe. 2018. "Inference in Regression Discontinuity Designs with a Discrete Running Variable." *American Economic Review* 108(8): 2277–2304.
- Lee, David S. 2008. "Randomized Experiments from Non-Random Selection in U.S. House Elections." *Journal of Econometrics* 142(2): 675–697.
- Lee, David S. 2009. "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects." *Review of Economic Studies* 76(3): 1071–1102.
- Lee, David S. and Thomas Lemieux. 2010. "Regression Discontinuity Designs in Economics." *Journal of Economic Literature* 48(2): 281–355.
- Legewie, Joscha. 2016. "Racial Profiling and Use of Force in Police Stops: How Local Events Trigger Periods of Increased Discrimination." *American Journal of Sociology* 122(2): 379–424.

References (cont.)

- Londoño-Vélez, Juliana, Catherine Rodríguez, and Fabio Sánchez. 2020. "Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia." *American Economic Journal: Economic Policy* 12(2): 193–227.
- McCrary, Justin. 2008. "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics* 142(2): 698–714.
- Noack, Claudia, Tomasz Olma, and Christoph Rothe. 2026. "Flexible Covariate Adjustments in Regression Discontinuity Designs." *Journal of Econometrics* 257: 106298.
- Sekhon, Jasjeet S. and Rocío Titiunik. 2017. "On Interpreting the Regression Discontinuity Design as a Local Experiment." *Advances in Econometrics* 38: 1–28.